

Next-Generation IEEE 802.11 standards

Introduction to the IEEE 802.11 standards

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Introduction to IEEE 802.11

Contents

- Introduction on Wi-Fi
- Traffic volumes and use cases of Wi-Fi
- History and evolution of Wi-Fi
- Standardization process



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Introduction on Wi-Fi

(Global) value of Wi-Fi

(Sep 2020) USA Internet Demand Still High

+73%

Average devices active online at home during working hours (vs. Jan-2020) ¹

+40%

Average household broadband usage per month (YOY vs. Sep-2019) ²

+31%

Peak upstream (videoconferencing) broadband traffic (vs. Mar-2020) ³

USA Internet Connectivity's Importance Will Continue to Increase

+84%

of USA businesses will increase WFH capacity post pandemic ⁵

\$186B

(\$2T) USA CARES Act funding available for telehealth & education remote learning ⁶

Wi-Fi's Role Is Rapidly Growing Globally

51%

of 2022 IP traffic is from Wi-Fi devices ⁴

59%

of 2022 mobile data offloaded to Wi-Fi ⁴

Wi-Fi Products Leverage the Latest Technologies to Support Evolving User Needs

Wi-Fi 6

Today

3x faster speeds ⁷
75% lower latency ⁸
4x device capacity ⁹

Wi-Fi 6E

~2021

Broad gigabit speed enabling
Reduced interference
Improved responsiveness

Wi-Fi 7

~2024

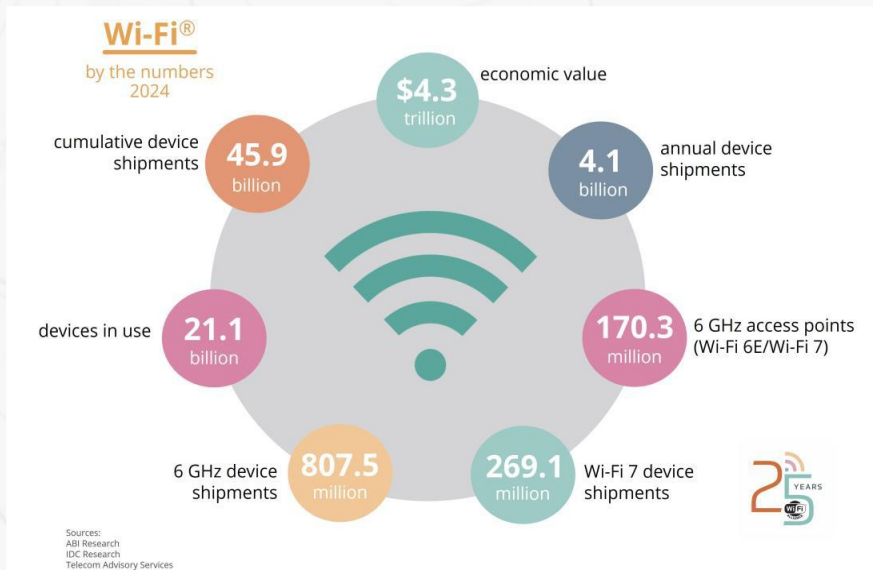
Nearly 5x faster speeds ¹⁰
Ultra-low latency
Enhanced reliability



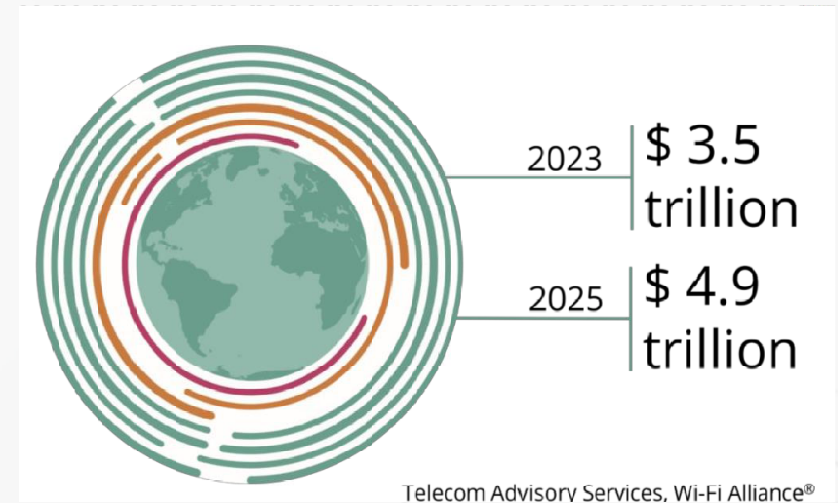
Introduction on Wi-Fi

(Global) value of Wi-Fi

Wi-Fi by the numbers¹¹



Global Value of Wi-Fi¹¹



¹¹Wi-Fi Alliance® and Telecom Advisory Services for Wi-Fi Alliance®, 2024 (<https://www.wi-fi.org/discover-wi-fi/value-wi-fi>)

Introduction on Wi-Fi

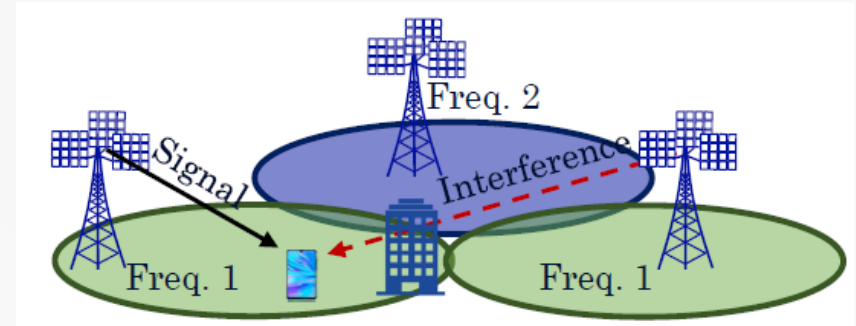
Cellular vs. Wi-Fi

Mobile cellular networks:

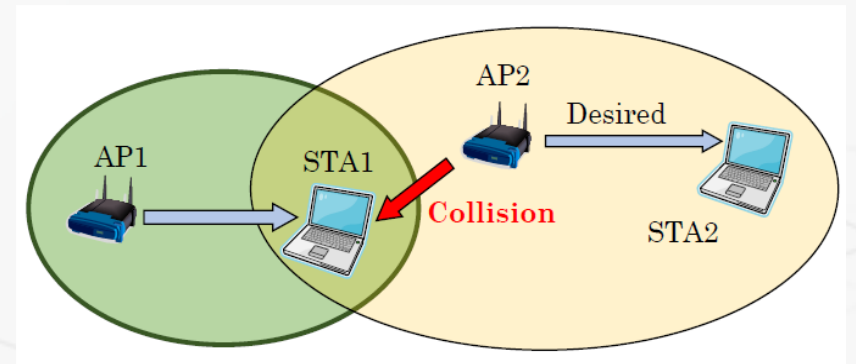
- Licensed spectrum allocated to a single operator
- Interference is under control via cell planning and radio resource management

Wi-Fi networks:

- Unlicensed spectrum (ISM bands), anyone can deploy a Wi-Fi network
- Subject to uncontrolled interference
- Packet collisions avoided via Distributed Coordination Function (DCF), comprising:
 - CSMA/CA + binary exponential backoff
 - Physical carrier sensing
 - Virtual carrier sensing: RTS/CTS, NAV, IFS
 - Auto repeat request (ARQ)



Cellular network with interference under control



Wi-Fi network with uncontrolled interference

Source: "Where Wi-Fi meets Ultra-High Reliability", IEEE Meditcom, 2024.

Introduction on Wi-Fi

Wi-Fi generations



Wi-Fi 4

Based on IEEE 802.11n

2.4 and 5 GHz band operation

Up to 4x4 MIMO, 40 MHz channels, 64-QAM

Max data rate: ~600 Mbps

Launched in 2007



Wi-Fi 5

Based on IEEE 802.11ac

5 GHz band operation

Up to 8x8 MIMO, DL MU-MIMO, 80 MHz and 160 MHz channels, 256-QAM

Max data rate: ~7 Gbps

Launched June/2013



Wi-Fi 6

Based on IEEE 802.11ax

2.4, 5, 6 GHz band operation

OFDMA, UL and DL MU-MIMO, target wake time (TWT), 1024-QAM

Best-in-class WPA3 security

Max data rate: 9.6 Gbps

Launched Aug/2019

Wi-Fi 6E
Extends Wi-Fi 6 with 6 GHz band operation
Launched Jan/2021



Wi-Fi 7

Based on IEEE 802.11be

2.4, 5, 6 GHz band operation

320MHz Channel Bandwidth

8 Spatial Streams, 4096-QAM

Multi-link Operation (MLO), Improved Target Wake Time

Improved Puncturing Transmission

Max data rate: 23 Gbps (one link), 35Gbps (multi-link)

Launched Dec/2023

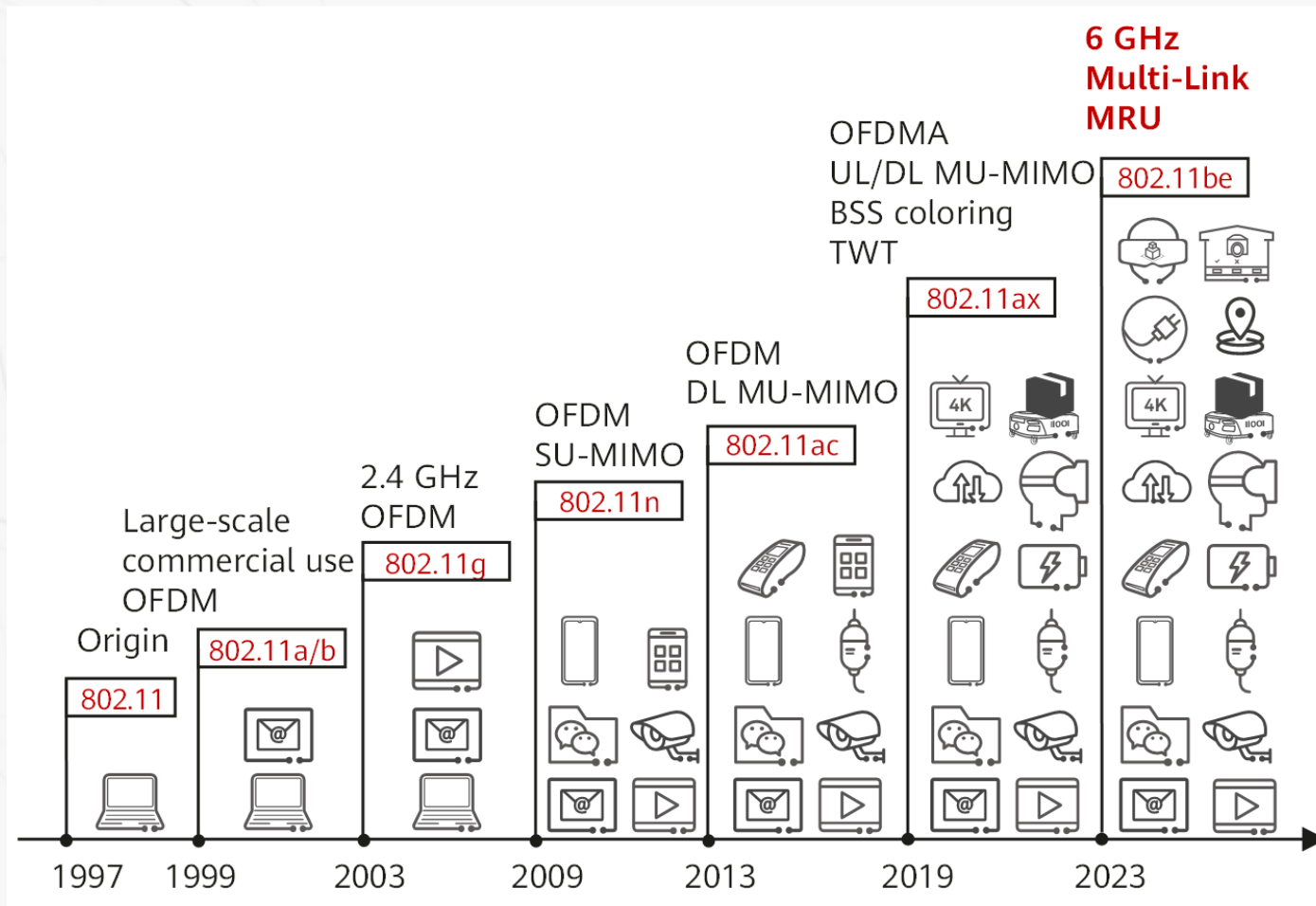
Source: "Wi-Fi Sensing: Use Cases, Standard Technologies, and Ecosystem Enablement", Intel, 2024.

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Introduction on Wi-Fi

IEEE 802.11 standard evolution

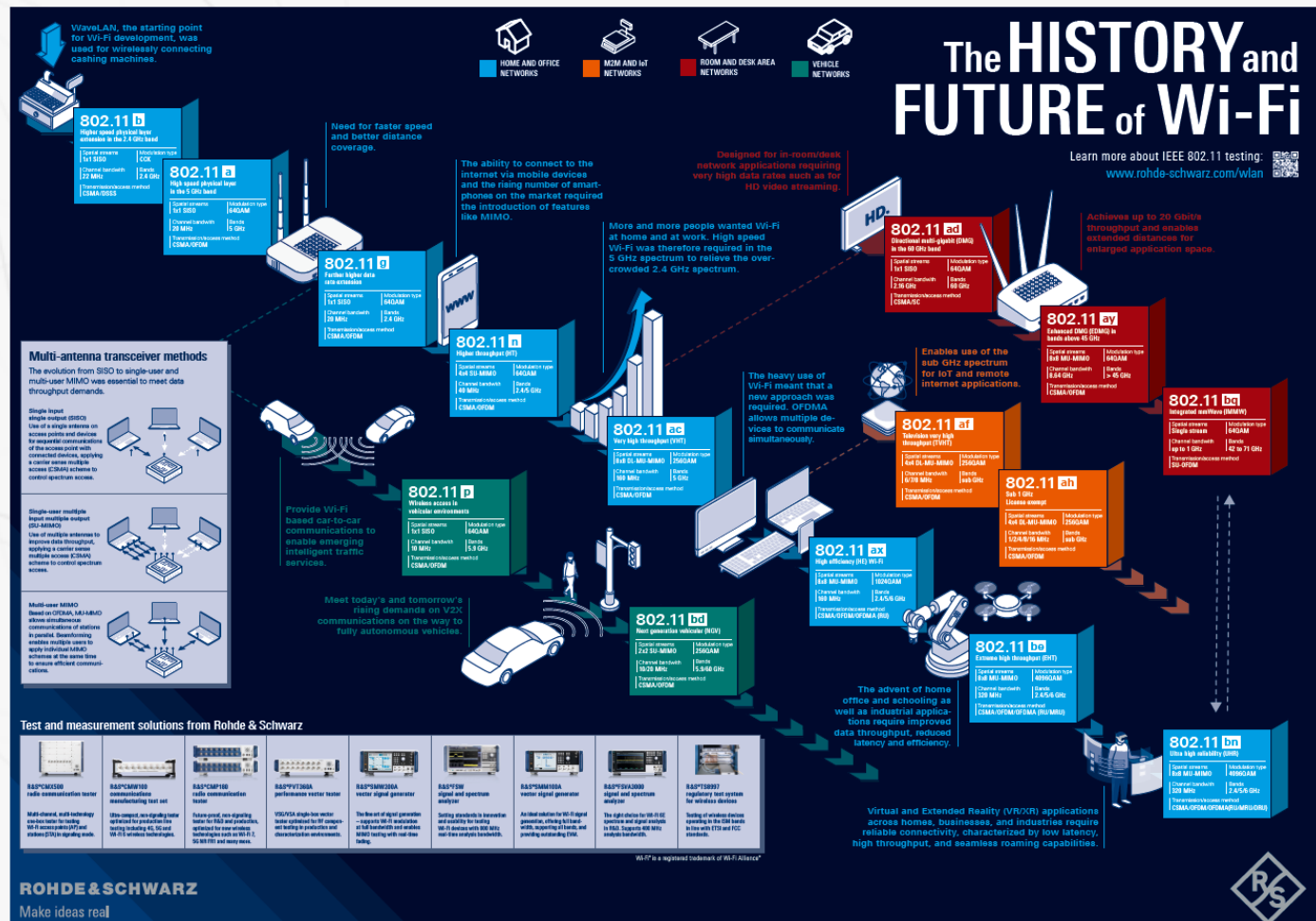


Source: "Wi-Fi 7", Huawei, 2025.



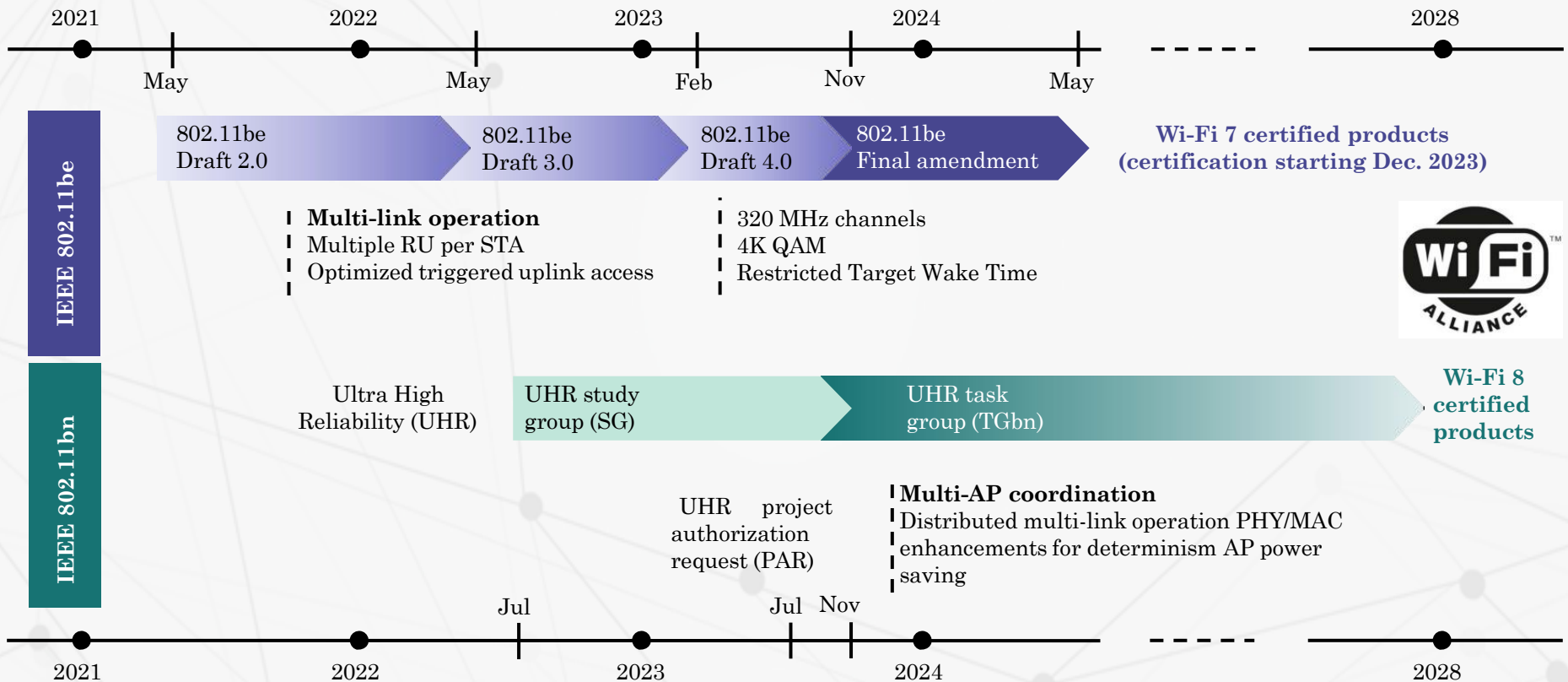
Introduction on Wi-Fi

History and future of Wi-Fi



Introduction on Wi-Fi

Ongoing standardization



Source: "Where Wi-Fi meets Ultra-High Reliability", IEEE Meditcom, 2024.



Introduction to IEEE 802.11

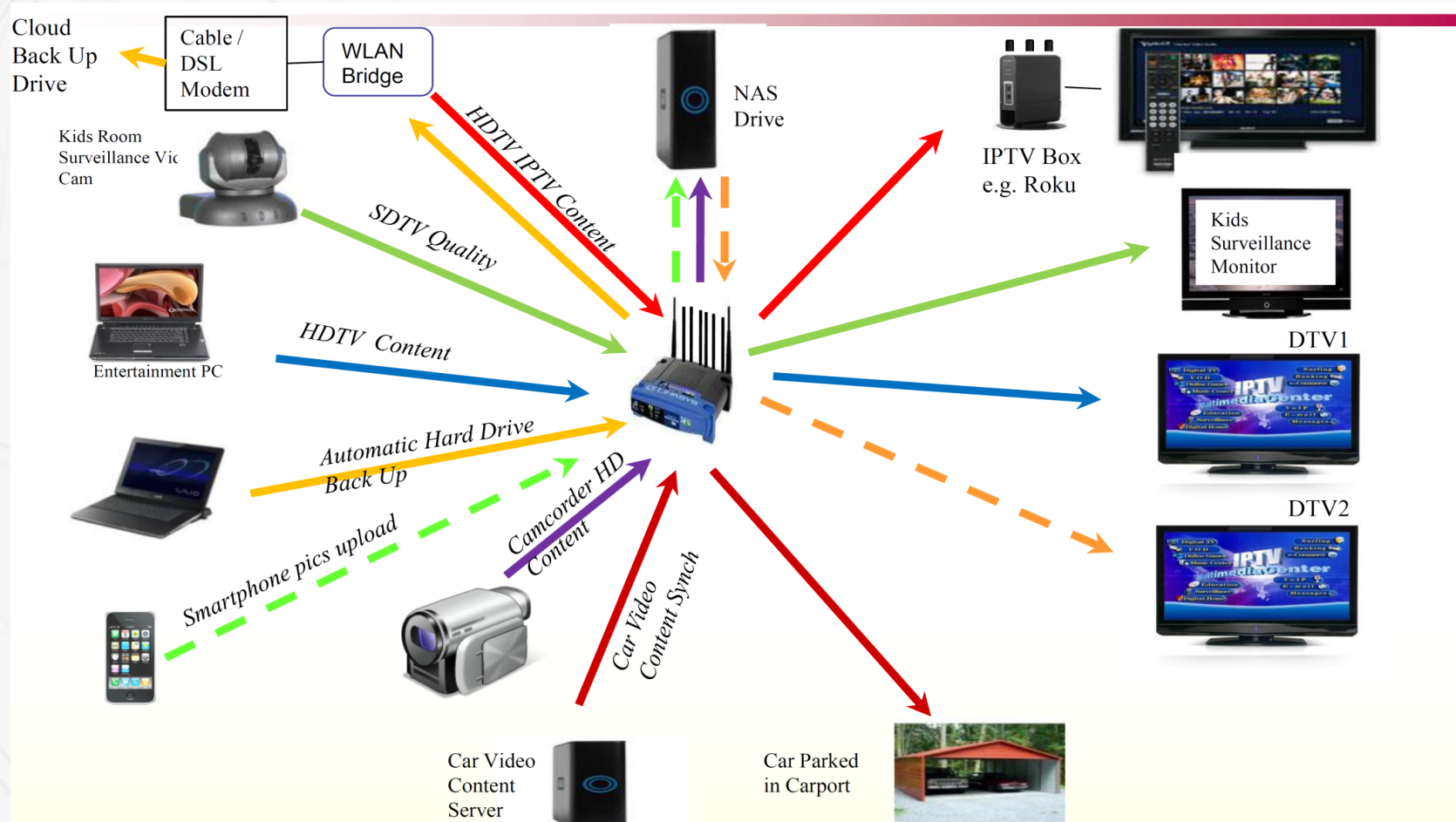
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Traffic volumes and use cases of Wi-Fi

Use cases from the past (how many years ago?)



Traffic volumes and use cases of Wi-Fi

Use cases from the past (how many years ago?)

Instant Wireless Sync

- IP-based P2P applications
- Using I/O PAL



Kiosk Sync & Data Exchange



Wireless Display

- HD streams over HDMI or DP using A/V PAL
- CE, PE and HH usages



Cordless Computing

- Combination of Wireless display using A/V PAL, sync and I/O using I/O PAL



Distributed Peripherals



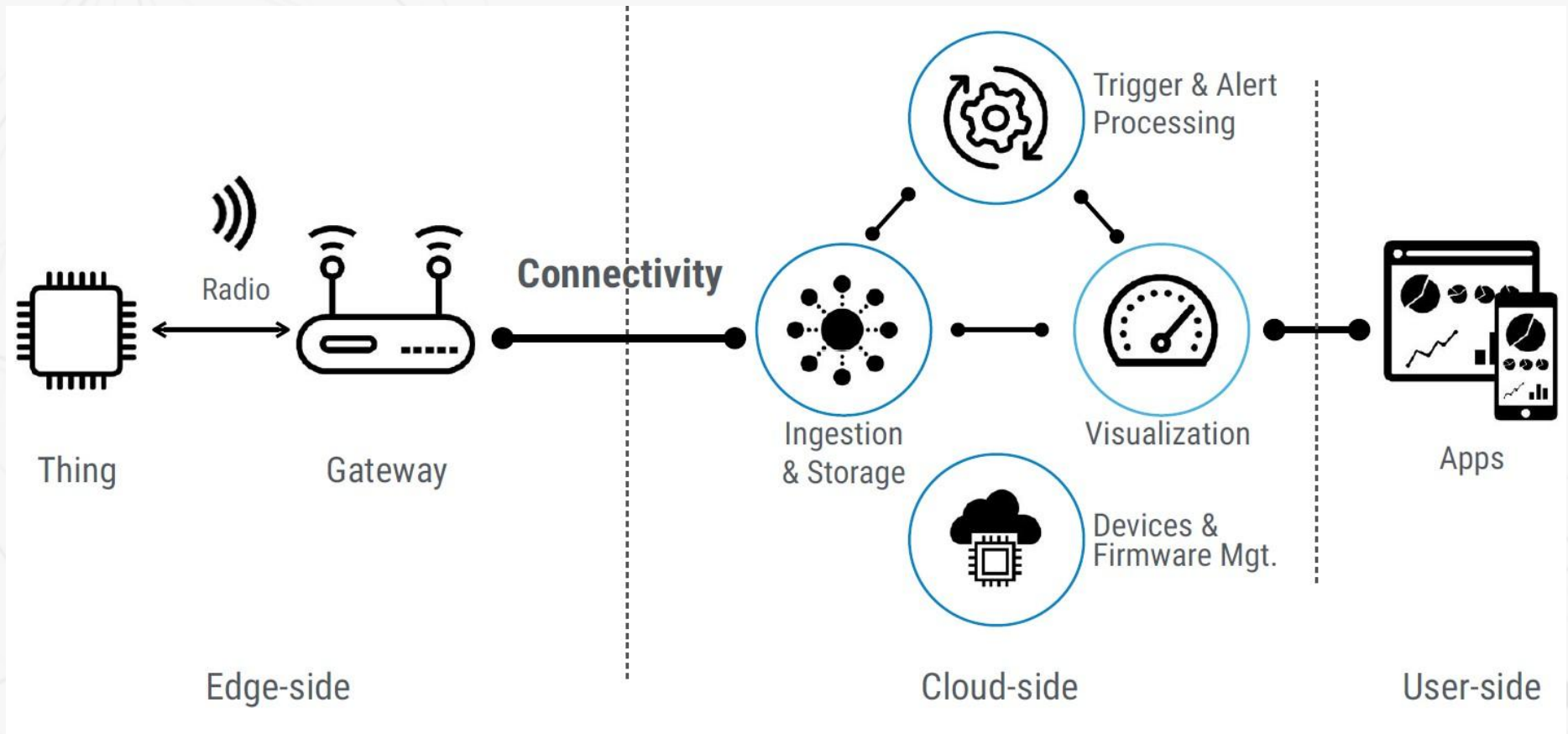
Internet Access

- Using native Wi-Fi, 802.11ad support



Traffic volumes and use cases of Wi-Fi

Internet of Things (IoT) architecture



Traffic volumes and use cases of Wi-Fi

Verticals

Smart
Wearables



Smart
Home



Smart
City



Smart
Agriculture



Connected
Car



Health
Care



Industry
Automation



Smart
Energy

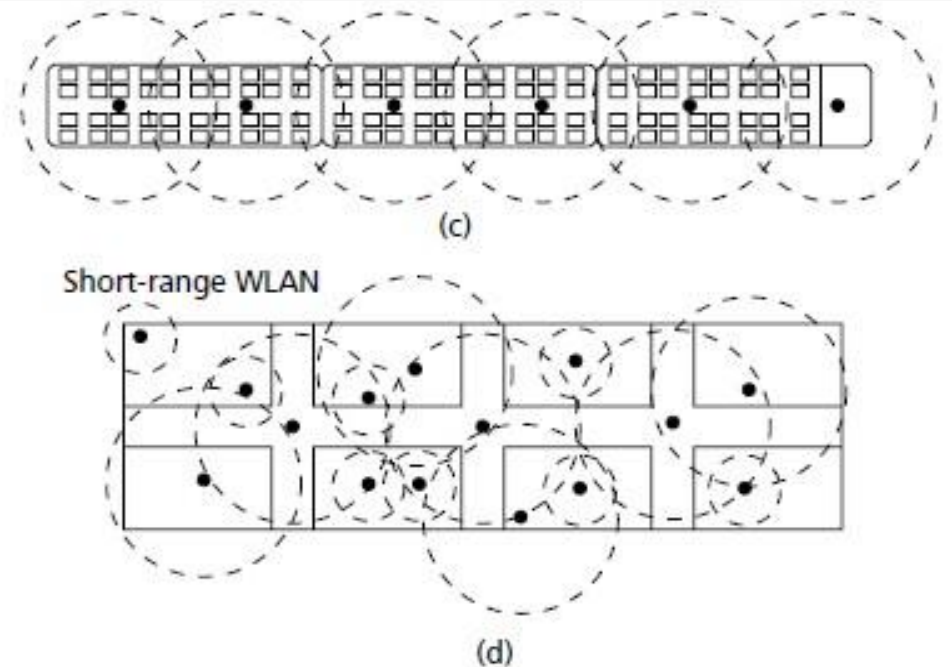
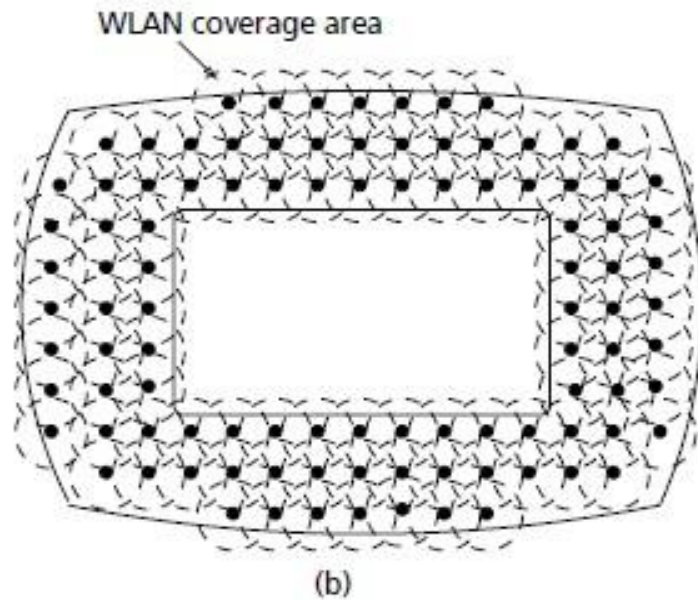


Source: Keysight Technologies



Traffic volumes and use cases of Wi-Fi

Very dense environments



Traffic volumes and use cases of Wi-Fi

Vehicle-to-everything (V2X)



Traffic volumes and use cases of Wi-Fi

Moving to 2030



Source: "Wi-Fi Sensing: Use Cases, Standard Technologies, and Ecosystem Enablement", Intel, 2024.

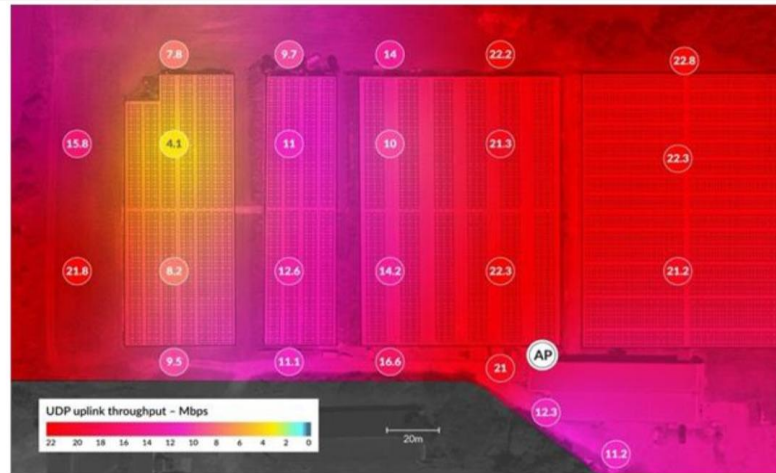
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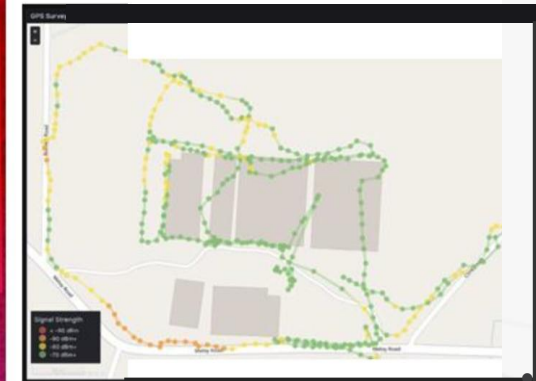
Traffic volumes and use cases of Wi-Fi

Smart farm

- A single AP covered the multi-acre farm, including multiple greenhouses and a small shop.



Green House Coverage



Whole Farm Coverage

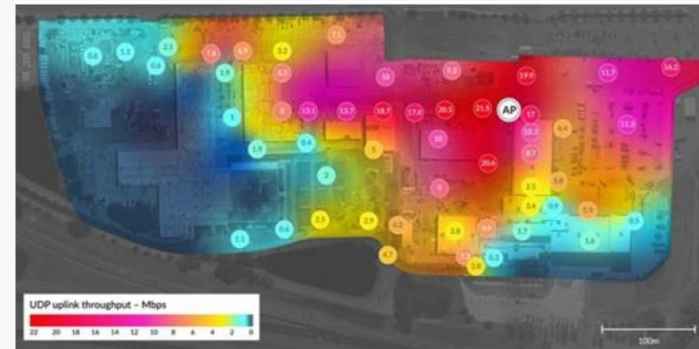
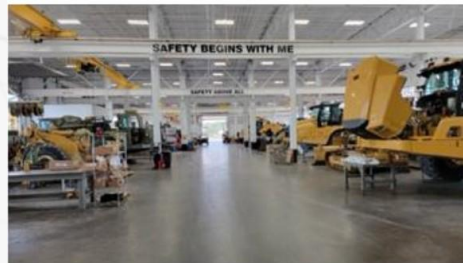
Source: "IEEE 802.11 Standards development enables innovation and productivity enhancements to build economic value", IEEE 802, 2024.



Traffic volumes and use cases of Wi-Fi

Smart industrial complex

- Two AP are able to cover the entire 1 km by 300m industrial site, with multiple high-bay workshops and large heavy equipment moving around the site.



Source: "IEEE 802.11 Standards development enables innovation and productivity enhancements to build economic value", IEEE 802, 2024.



Traffic volumes and use cases of Wi-Fi

Industrial environments



Introduction to IEEE 802.11

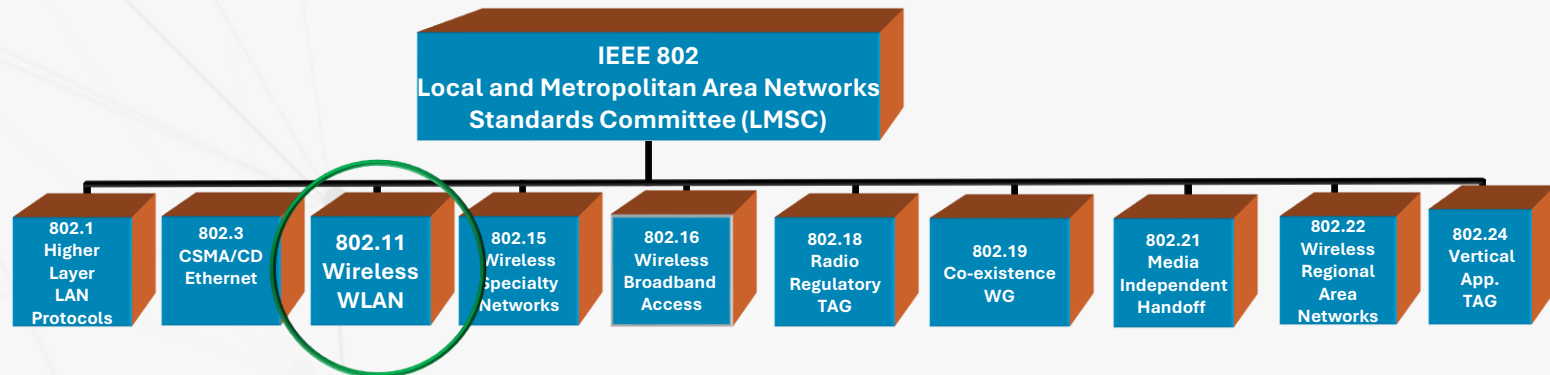
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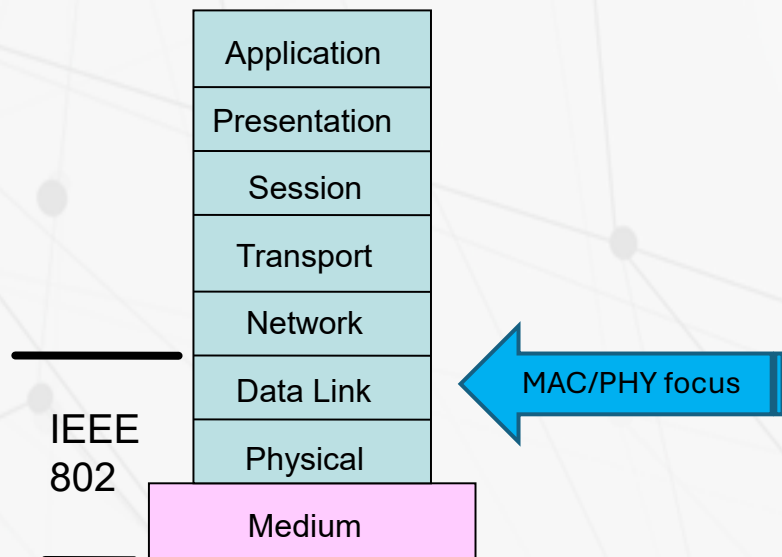
History and evolution of Wi-Fi

The IEEE 802.11 Working Group is one of the most active WGs in 802



OSI Reference Model

IEEE 802.11 WG Voting Members: 600+

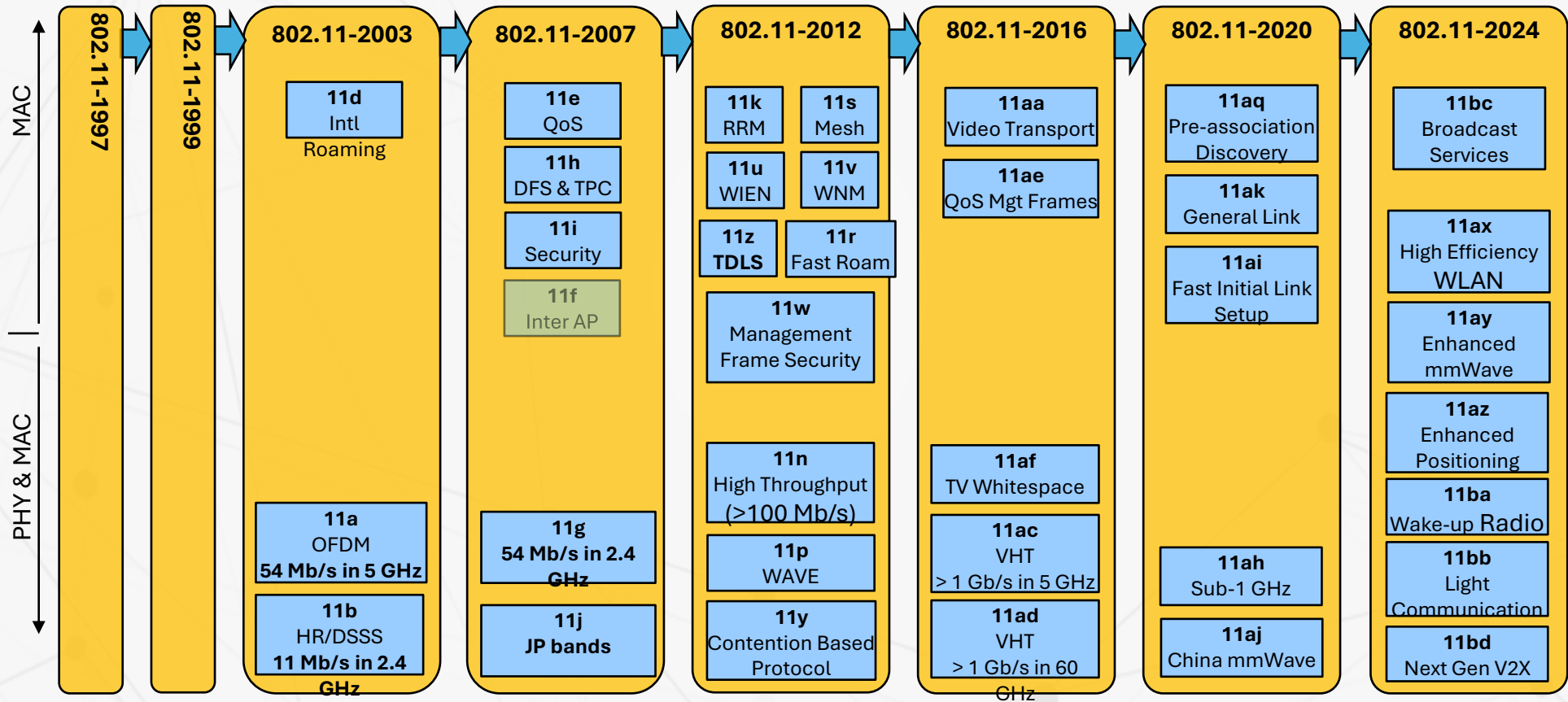


Source: "IEEE 802.11 Overview and Amendments under development", IEEE 802, 2025.



History and evolution of Wi-Fi

Development of the IEEE 802.11 Standard is ongoing since 1997



Source: "IEEE 802.11 Overview and Amendments under development", IEEE 802, 2025.



History and evolution of Wi-Fi

Completed standards

IEEE 802.11-2020 Revision project

Includes the following amendments:

IEEE 802.11ah – Operating in license-exempt bands below 1 GHz (excl. TV bands)

IEEE 802.11ai – Fast initial link set-up

IEEE 802.11aj – Enable 802.11ad operation in the Chinese 59-64 GHz frequency band.

IEEE 802.11ak – General Link

IEEE 802.11aq – Pre-association Service Discovery

Source: “IEEE 802.11 Overview and Completed Amendments”, IEEE 802, 2025.



History and evolution of Wi-Fi

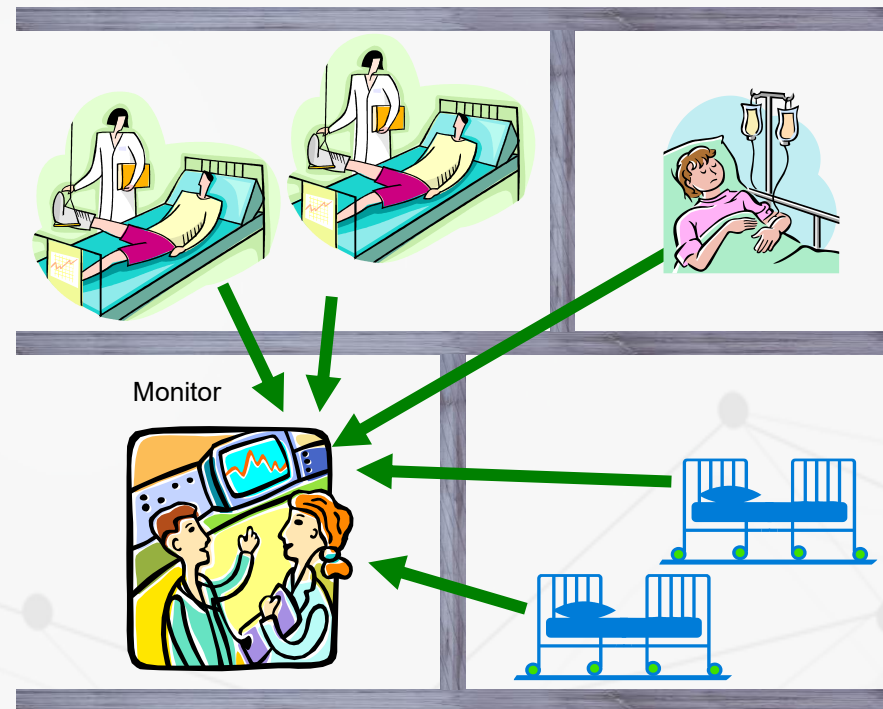
802.11ah defines OFDM PHY and MAC operating in license-exempt bands below 1 GHz (excl. TV bands)

Define OFDM PHY and MAC operating in license-exempt bands below 1 GHz (excl. TV bands)

- Extended range
- Power efficiency
- Large number of devices

Potential applications

- Internet of everything (IoT)
- Smart Grid
- Healthcare
- Smart Appliances
- Wearable consumer electronics



History and evolution of Wi-Fi

802.11ai defines fast initial link set-up methods

This amendment defines mechanisms that provide IEEE 802.11 networks with fast initial link set-up methods which do not degrade the security currently offered by Robust Security Network Association (RSNA) already defined in IEEE 802.11.

The project's primary need came from an environment where mobile users are constantly entering and leaving the coverage area of an existing extended service set (ESS).

- scale with a high number of users simultaneously entering an ESS
- minimize the time spent within the initial link set-up phase
- securely provide initial authentication.



History and evolution of Wi-Fi

802.11aj defines modifications to 802.11ad to enable operation in the Chinese 59-64 GHz frequency band

This amendment defines modifications to the IEEE P802.11ad Physical (PHY) layer and the Medium Access Control (MAC) layer to enable operation in the Chinese 59-64 GHz frequency band.

The amendment maintains backward compatibility with 802.11ad when it operates in the 59-64 GHz frequency band.

The amendment also defines modifications to the PHY and MAC layers to enable the operation in the Chinese 45 GHz frequency band. The amendment maintains the 802.11 user experience.



History and evolution of Wi-Fi

802.11ak enables an 802.11 connection to be used as a through link in a general network

This amendment enables an 802.11 connection to be used as a through link in a general network, not just as a connection to an end station at the edge of a network.

Fully general mixed 802.11 and wired plug and play in the home.

Data Center top-of-rack to top-of-rack connections for overflow traffic.

Industrial and Enterprise network use.



History and evolution of Wi-Fi

802.11aq enables the delivery of pre-association Service Discovery information

A typical use case is printer discovery in a hotel.

Pre-association protocol designed to discover services on a WLAN.



History and evolution of Wi-Fi

Completed standards

IEEE Std 802.11-2024 Revision project

Includes the following amendments:

IEEE Std 802.11ax-2021 – Increased throughput & efficiency in 2.4, 5 (and 6) GHz bands

IEEE Std 802.11ay-2021 – Support for 20 Gbps in 60 GHz band

IEEE Std 802.11az-2022 – 2nd generation positioning features

IEEE Std 802.11ba-2021 – Wake up radio. Low power IoT applications

IEEE Std 802.11bb-2023 – Light Communications

IEEE Std 802.11bc-2023 – Enhanced Broadcast Service

IEEE Std 802.11bd-2022 – Enhancements for Next Generation V2X

IEEE Std 802.11be-2024 – Enhancements for extremely high throughput (EHT)

IEEE Std 802.11bh-2024 – Operation with Randomized and Changing MAC Addresses

Source: “IEEE 802.11 Overview and Completed Amendments”, IEEE 802, 2025.



History and evolution of Wi-Fi

802.11ax is focused on improving performance in dense environments

Existing 802.11 WLAN systems serve dense deployments. IEEE 802.11ax further improves performance of WLAN deployments in dense scenarios:

- Targeting at least 4x improvement in the per-STA throughput compared to 802.11n and 802.11ac.
- Improved efficiency through spatial (MU MIMO) and frequency (OFDMA) multiplexing.

Dense scenarios are characterized by large number of access points and large number of associated STAs deployed in geographical limited region (e.g. a stadium or an airport).



History and evolution of Wi-Fi

802.11ay is defining next generation 60 GHz: increased throughput and range

20Gbps+ rates are defined

License- Exempt bands above 45 Gbps

Completion in 2020

Use Cases:

- Ultra-Short Range
- 8K UHD - Smart Home
- AR/VR and wearables
- Data Center Inter Rack connectivity
- Video / Mass-Data distribution
- Mobile Offloading and MBO
- Mobile Fronthauling
- Wireless Backhauling (w. multi-hop)
- Office Docking
- Fixed Wireless

Key additions :

- SU/ MU MIMO, up to 8 spatial streams
- Channel bonding
- Channel aggregation
- Non-uniform constellation modulation
- Advanced power saving features



History and evolution of Wi-Fi

802.11az Next Generation Positioning

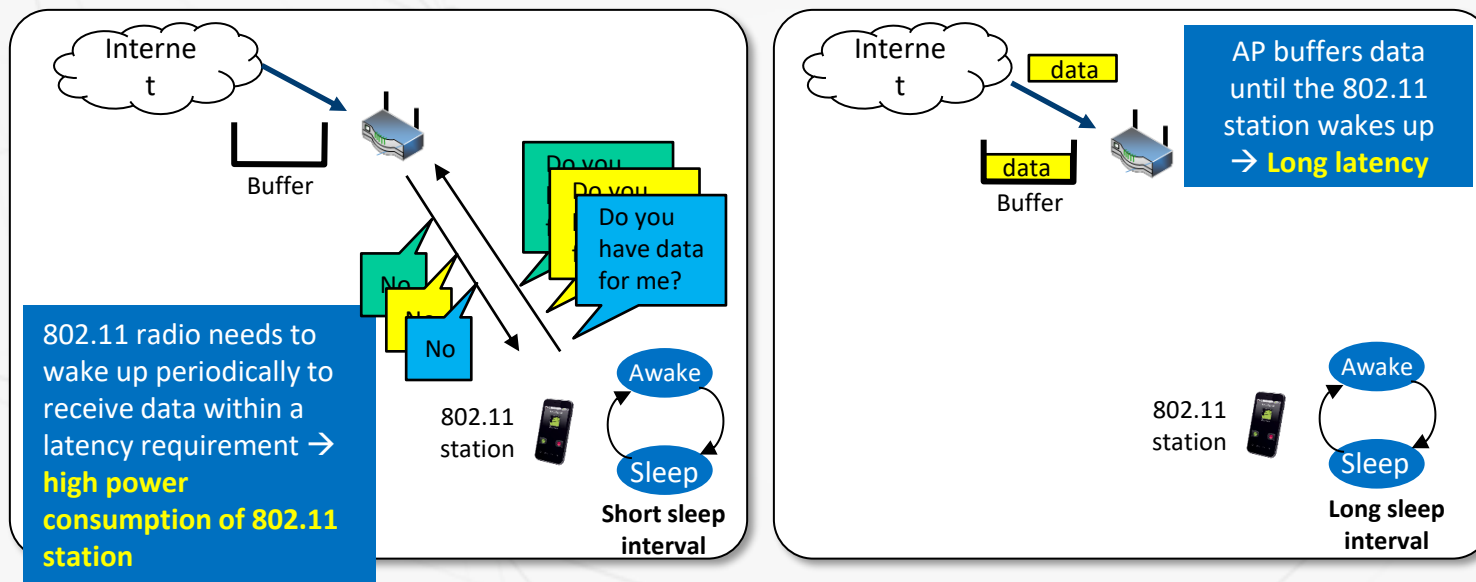
The P802.11az project Next Generation Positioning extends accurate IEEE Std 802.11-2016 Fine Timing Measurement capabilities:

- Accurate indoor Navigation (sub 1m and into the $<0.1\text{m}$ domain).
- Enables self-locating networks for easy, fast and cost efficient WLAN deployment for navigation and 6GHz AFC operation.
- Secured (authenticated and private) positioning – open my car with my smartphone, position aware services (money withdrawal).
- Unlock computer with a wearable device, adapt TV content to audience presence.
- Location based link adaptation for home usages (connect to best AP).
- Navigate in extremely dense environments (stadium/airport scenarios).



History and evolution of Wi-Fi

802.11ba improves energy efficiency of stations and maintains low latency



History and evolution of Wi-Fi

802.11bb Light Communications

Ability to re-use existing HT, VHT and HE PHY/MAC.

Use Cases:

- Industrial wireless applications
- Medical environments
- Enterprise
- Home
- Backhaul
- Vehicle to Vehicle Communication
- Underwater Communication
- Gas Pipeline Communication

Key additions :

- Uplink and downlink operations in 800 nm to 1000 nm band
- Minimum single-link throughput of 10 Mb/s
- Interoperability among solid state light sources with different modulation bandwidths.



History and evolution of Wi-Fi

802.11bc is defining Enhanced Broadcast Services

Enhanced Broadcast Services (eBCS) defines broadcast service enhancements within an 802.11-based network.

Client end devices broadcast information to an AP, e.g. in an IoT environment, to other STAs so that any of the receiving APs act as an access node to the Internet.

Broadcast Downlink

- Provides enhanced Broadcast Services (eBCS) of data (e.g. videos) to a large number of densely located STAs.
- These STAs may be associated, or un-associated with the AP or may be low-cost STAs that are receive only.



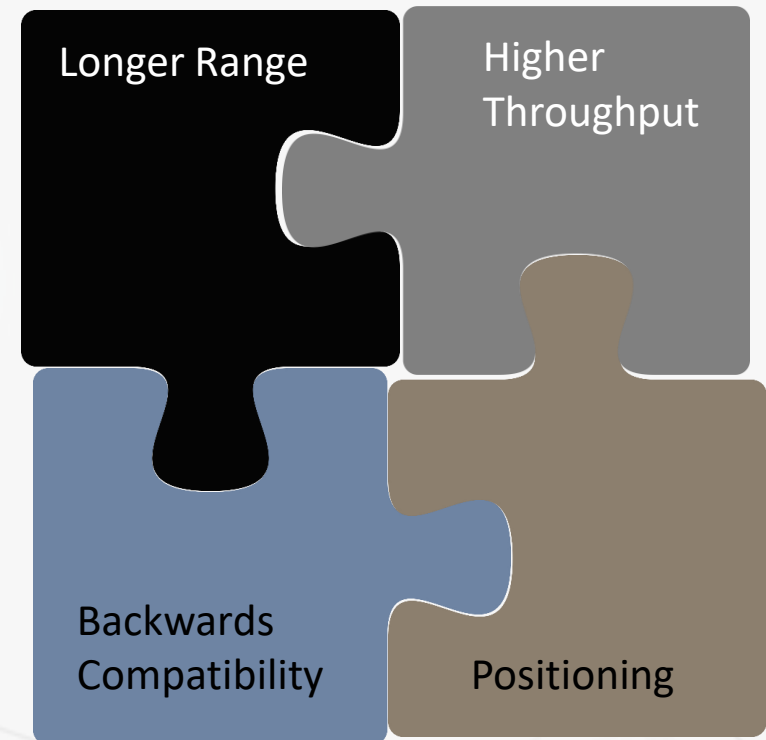
History and evolution of Wi-Fi

802.11bd defines an evolution of 802.11p for Vehicle to Anything (V2X)

IEEE 802.11p is largely based on 802.11a.

IEEE 802.11bd defines MAC/PHY enhancements from 802.11n, ac, ax, to provide a backwards compatible next generation V2X protocol.

- Higher Throughput
- Longer Range
- Support for Positioning
- Backward Compatibility



History and evolution of Wi-Fi

802.11bd: Next Generation V2X Use Cases

5.9 GHz band mainly, and optionally 60 GHz

V2X Use Cases:

- Support all defined DSRC/802.11p use cases, including Basic safety message (safety, range, backward compatibility, fairness)
- Sensor sharing (throughput)
- Multi-channel operation (safety channel + other channels)
- Infrastructure applications (throughput)
- Vehicular positioning & location (LoS and NLoS positioning accuracy)
- Automated driving assistance (safety, throughput)
- Aerial vehicle IT application (video)
- Train to train (high speed)
- Vehicle to train (high speed, long range)

Key additions (comparing to 802.11p):

- Higher throughput (2x)
- Longer range
- Support for positioning
- Backward compatibility with 11p



History and evolution of Wi-Fi

802.11be Extremely High Throughput amendment builds on 802.11ax, including 6GHz support

Extremely High Throughput (EHT)

Operation in 2.4 GHz, 5 GHz, and 6 GHz bands

Higher throughput – Project goal of at least 30 Gbps; expect 40+Gbps with 320MHz channels, 4096 QAM

Support for low latency communications

Use Cases:

- Home, enterprise, industrial, IoT
- Outdoor
- AR/VR, wireless gaming
- 4K and 8K video streaming
- Remote office
- Cloud/edge computing
- Video calling and conferencing



History and evolution of Wi-Fi

Market demands and new technology drive IEEE 802.11 innovation

Demand for throughput

- Continuing exponential demand for throughput (*802.11be and 802.11bn*)
- Most (50-80%, depending on the country) of the world's mobile data is carried on 802.11 (Wi-Fi) devices

New usage models / features

- Dense deployments (*802.11be*), Indoor Location (*802.11az, 802.11bk*),
- Automotive (*802.11bd, AUTO*)
- WLAN Sensing (*802.11bf*)

Technical capabilities

- Millimetric (42-71 GHz) radios (*802.11bq*)

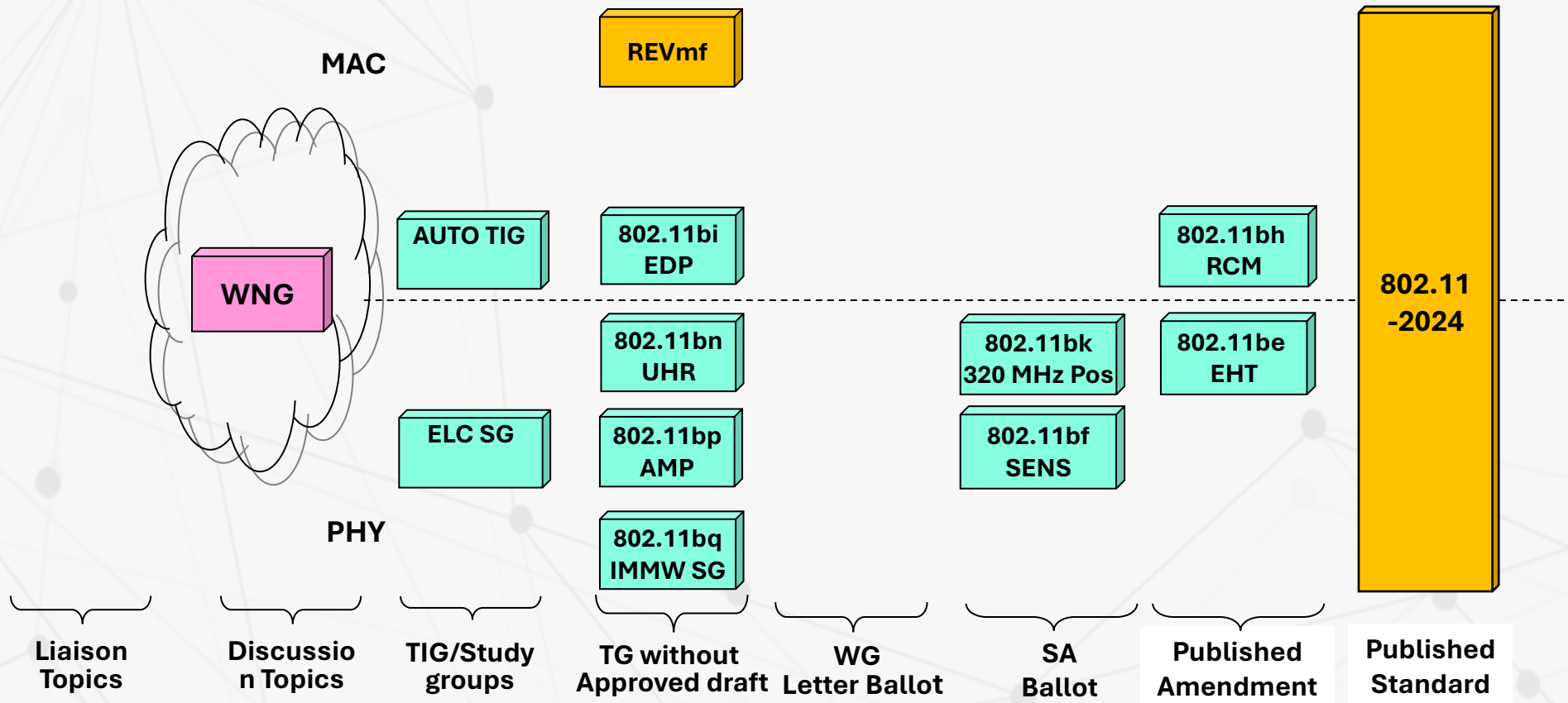
Changes to regulation

- 6 GHz (*802.11be*)
- Coexistence and radio performance rules (e.g., ETSI BRAN, ITU-R)



History and evolution of Wi-Fi

IEEE 802.11 standards pipeline (January 2025)



Source: "IEEE 802.11 Overview and Amendments under development", IEEE 802, 2025.



History and evolution of Wi-Fi

New IEEE 802.11 radio technologies

Under development to meet expanding market needs and leverage new technologies

802.11bf – WLAN Sensing

802.11bi – Enhanced Data Privacy

802.11bk – 320 MHz Positioning

802.11bn – Ultra High Reliability

802.11bp – Ambient Power for IOT

802.11bq – Integrated Milli Metric Wave

ELC – Enhanced Light Communication Study Group

AUTO – Automotive Topic Interest Group

AI / ML – Artificial Intelligence / Machine Learning Group



History and evolution of Wi-Fi

Wi-Fi standard evolution: 802.11bn is now under development

	Project	Industry Name	Defining features
Completed	802.11n High Throughput	Wi-Fi 4	Spatial multiplexing, 40 MHz channels, beamforming, A-MPDU
	802.11ac Very High Throughput	Wi-Fi 5	80 MHz & 160 MHz channels, beamforming that works Enabled broad support for 5 GHz band operation
	802.11ax High Efficiency	Wi-Fi 6 and 6E	Multi-user operation, 320 MHz channels, 6 GHz band operation
Recently approved	802.11be Extremely High Throughput	Wi-Fi 7	Multi-link operation (simultaneous use of multiple channels)
New	802.11bn Ultra-High Reliability	Wi-Fi 8	Lower latency, longer range, faster handover

Source: "IEEE 802.11 Overview and Amendments under development", IEEE 802, 2025.



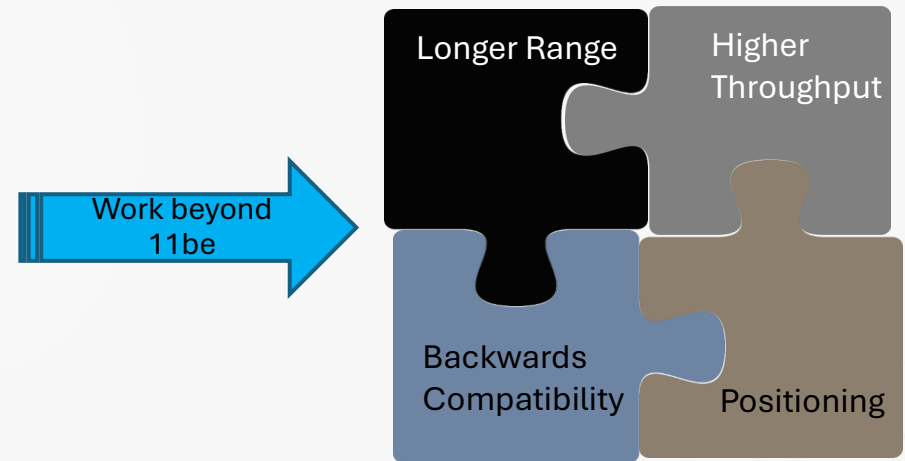
History and evolution of Wi-Fi

IEEE 802.11bn: Ultra High Reliability (UHR)

Expected to be the basis for Wi-Fi 8

Expected improvements:

- Reduce tail latency
- Reduce roaming latency by taking advantage of multi-link features
- Allow access on secondary channel while primary channel is busy
- AP power save
- Security enhancements, e.g., control frame protection
- Extend range by reducing sensitivity gap between client and AP
- Multi-AP coordination



History and evolution of Wi-Fi

802.11bf Sensing

IEEE 802.11bf is developing a protocol for environmental sensing

Measurements that can be used to monitor environmental conditions and changes (e.g., people movement, number of people present, room occupancy)

Built on sounding (beamforming) waveforms

WLAN sensing uses PHY and MAC features of IEEE 802.11 stations to obtain **measurements** that may be useful to estimate **features** of **objects** in an **area of interest**.

- Features = Range, velocity, angular, motion, presence or proximity, gesture, etc.
- Objects = Human, animal, etc.
- Area of interest = Room, car, enterprise, etc.



History and evolution of Wi-Fi

802.11bf Sensing

Use cases:



1. Smart home



2. Presence and proximity detection



3. Gesture recognition



4. Gaming control



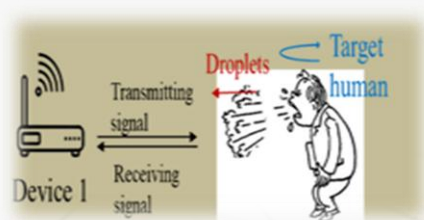
5. Vital signs / Liveness



6. Location in store



7. Audio with user tracking (Follow-me sound)



8. Sneeze sensing



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Standardization process

What standards are (in a wide sense)

The most general definition for a «standard» may be

«a widely agreed way of **doing something**»

.... where, depending on the specific area of application, “**doing something**” may be replaced by, e.g., “designing a product”, “building a process”, “implementing a procedure” or “delivering a service”.

«**Standard**» (i.e. agreed and common) ways of doing things bring lot of benefits; our technological world without «standards» simply would not work (or, at least, it would be harder to make it work)

Source: “Understanding ICT Standardization: Principles and Practice”, ETSI, 2021



Standardization process

What standards are? (various definitions)

Standards are published documents that establish specifications and procedures designed to maximize the reliability of the materials, products, methods, and/or services people use every day. Standards address a range of issues, including but not limited to various protocols to help maximize product functionality and compatibility, facilitate interoperability and support consumer safety and public health. (definition by IEEE Standards Association, IEEE-SA)

Standard is a document, established by consensus and approved by a recognized body, that provides, for common and repeated use, rules, guidelines or characteristics for activities or their results, aimed at the achievement of the optimum degree of order in a given context. (derived from ISO/IEC Guide 2:1996, definition 3.2)

Standards are produced by a variety of organizations, including scientific and professional associations, industry and trade organizations and governments.



Standardization process

Why Standards?

- **Interoperability:** The paramount benefit. Standards ensure that products, systems, and services from different manufacturers can communicate and function together seamlessly (e.g., charging your phone with any USB-C charger, connecting to any Wi-Fi network).
- **Market efficiency:** Reduces market fragmentation and creates larger, more predictable markets, benefiting both producers and consumers (more choice, lower prices).
- **Innovation catalyst:** Provides a stable base layer, allowing companies to focus their innovation efforts on value-added features and applications built upon common standards.
- **Quality and reliability:** Standards often include performance metrics, testing procedures, and best practices, leading to higher quality and more reliable products and services.
- **Safety and health:** Many standards directly address safety requirements, environmental impacts, and ergonomic considerations, protecting users and the environment.
- **Facilitating trade:** International standards remove technical barriers, simplifying global trade and commerce.



Standardization process

Different types of standards

According to the development process (standardization)

De facto standards, or standards in actuality, are adopted widely by an industry and its customers. These standards arise when a critical mass simply likes them well enough to collectively use them.

De jure standards are produced by devoted organizations, called **Standards Development Organizations (SDOs)**. SDOs are organizations whose purpose is to develop standards and that put in place formal well-defined procedures to guarantee a fair development process.



Standardization process

What are SDOs?

An **SDO** is an organization whose primary activities include developing, coordinating, promulgating, revising, amending, or interpreting voluntary consensus standards.

- Purpose: SDOs bring together diverse stakeholders – including industry experts, academics, government officials, regulators, and consumer representatives – to collaborate on technical specifications.
- Voluntary consensus: The standards developed are typically voluntary, meaning their adoption is not legally mandated (though they may be incorporated into regulations). They are also "consensus-based," signifying general agreement among participants.

SDOs are crucial for:

- Ensuring interoperability between different products and systems.
- Fostering innovation by providing stable technical foundations.
- Promoting market growth and reducing technical barriers to trade.
- Enhancing safety, quality, and environmental sustainability.



Standardization process

What standards are NOT & classification of SDOs

Standards are NOT regulations

- While conformity with standards is voluntary, regulations are compulsory

Standards are NOT a set of thorough design rules

- Standards are aimed at defining a minimum set of requirements for an item (product, service, process, etc.) in order to make it meet certain well-defined objectives

Standards are voluntary NOT compulsory

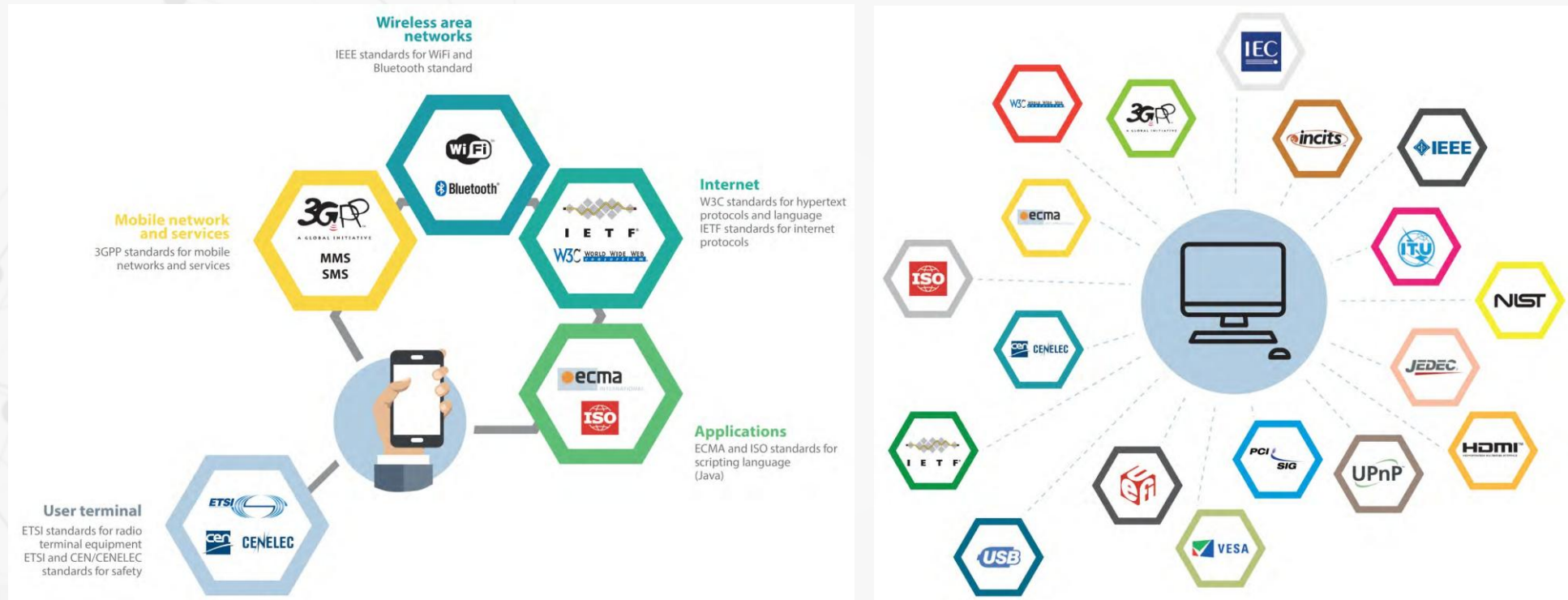
Standardization landscape includes multiple SDOs that may differ in:

- Geographical coverage
- Technical scope of activities (as per each SDO's statute)
- Level of recognition from regulatory or political organizations



Standardization process

Standards in everyday life



Source: “Understanding ICT Standardization: Principles and Practice”, ETSI, 2021



Standardization process

Classification of SDOs

International SDOs

- These have members worldwide, which sometimes also include national or regional standard bodies, and their deliverables have worldwide coverage.



Regional SDOs

- These have members (industries, academia and national SDOs) from countries that usually share, or are interested in promoting common practices and regulations.



National SDOs

- National SDOs operate at the single country level and issue country-specific standards; they often collaborate with International and Regional SDOs.



Standardization process

Classification of SDOs - Examples of scope of activities

Organization	Typical technical scope of activity
ITU	Interoperable telecom specifications incl. architecture, services, protocols, addressing / numbering plans
ISO	ICT architecture (OSI model) services, protocols incl. application protocols
IEC	Electrotechnical standards, incl. connectors, electrical safety and tests
ETSI	Standards for ICT-enabled systems, applications and services
CEN	Household appliances, Intelligent Transportation and Mobility, Smart Grids and Smart Metering, Cybersecurity, Blockchains
CENELEC	Electrotechnical standards, incl. connectors, electrical safety and tests, ECM
IEEE	All LAN specifications: IEEE 802.xx, including cabled LANs, Token Ring and Bus, Wireless LANs WLAN, e.g. WiFi)
IETF	All internet related specifications including protocols, generic applications, addressing rules (IP, url)

Source: “Understanding ICT Standardization: Principles and Practice”, ETSI, 2021



Standardization process

Core principles of standardization by SDOs

While specific procedures vary, most reputable SDOs adhere to common principles:

- **Consensus:** Decisions are based on general agreement among participants, ensuring that no single interest group can dominate. It implies reconciliation of conflicting arguments.
- **Openness:** The process is transparent, and all interested and qualified parties are eligible to participate, ensuring a broad range of perspectives.
- **Due process:** A formal, documented procedure is followed at every stage, providing fairness, transparency, and a path for appeals.
- **Technical merit:** Standards must be technically sound, relevant to market needs, and technologically feasible.
- **Global relevance:** Standards should aim for worldwide applicability and not create unnecessary trade barriers.
- **Intellectual Property Rights (IPR) policy:** Fair, reasonable, and non-discriminatory (FRAND/RAND) licensing of essential patents is typically required.



Standardization process

The role of IEEE in standardization

- **IEEE (Institute of Electrical and Electronics Engineers)** is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity. It has over 400,000 members in over 160 countries.
- **Role in standardization:** IEEE is a leading developer of international standards in a broad range of industries, particularly in electronics, power, computing, and communications.
- **Impact:** IEEE standards are fundamental to the operation of countless technologies, from Wi-Fi and Ethernet to power grids and medical devices, enabling interoperability and driving innovation globally.
- **IEEE Standards Association (IEEE-SA)** is the organizational unit within IEEE responsible for managing the standardization process. It is composed of individual members and corporate organizations from around the world.



Standardization process

The standards development lifecycle

IEEE Comm. Society Standards Development Board (IEEE COM/SDB) *

- Sponsors standards in communications & networking

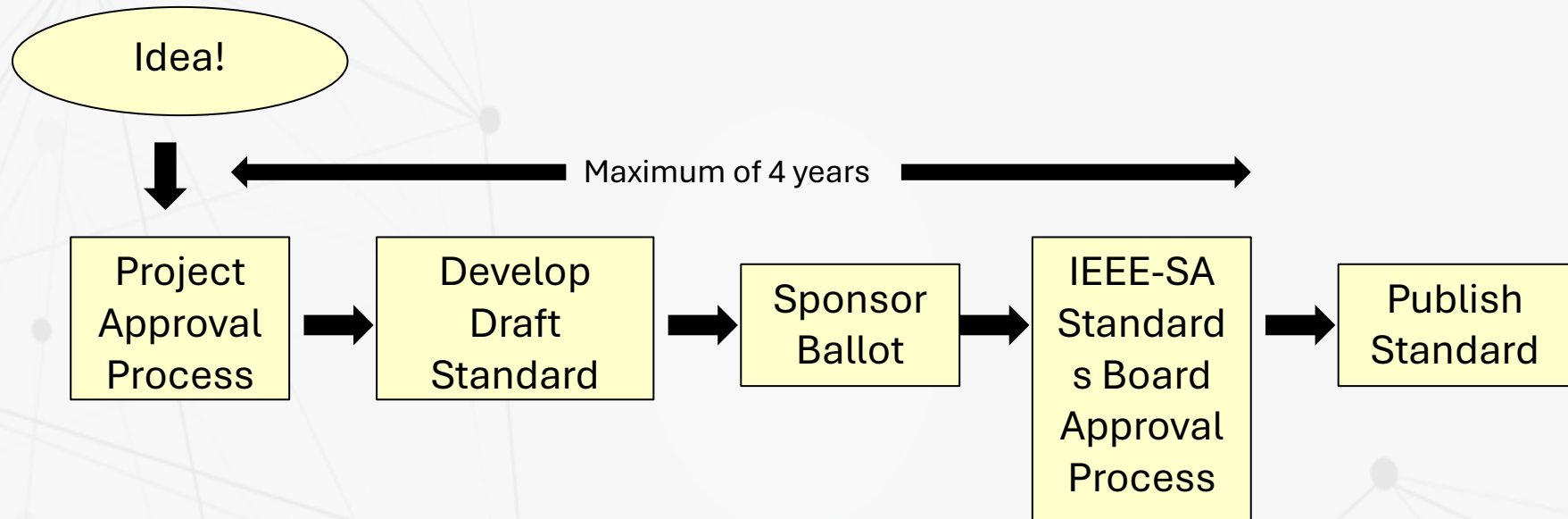


Source: “IEEE-SA quick reference guide standards development process” and <https://standards.ieee.org/about/policies/bylaws/> (IEEE-SA Standards Board)



Standardization process

Process flow

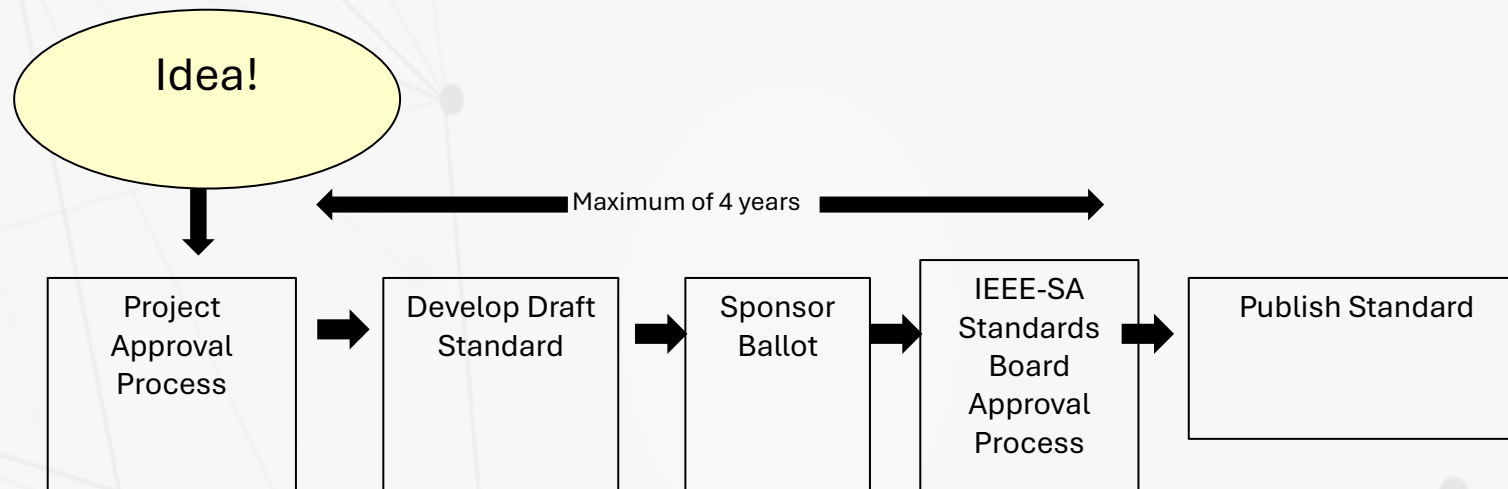


It is a very long and complicated process!



Standardization process

Getting Started



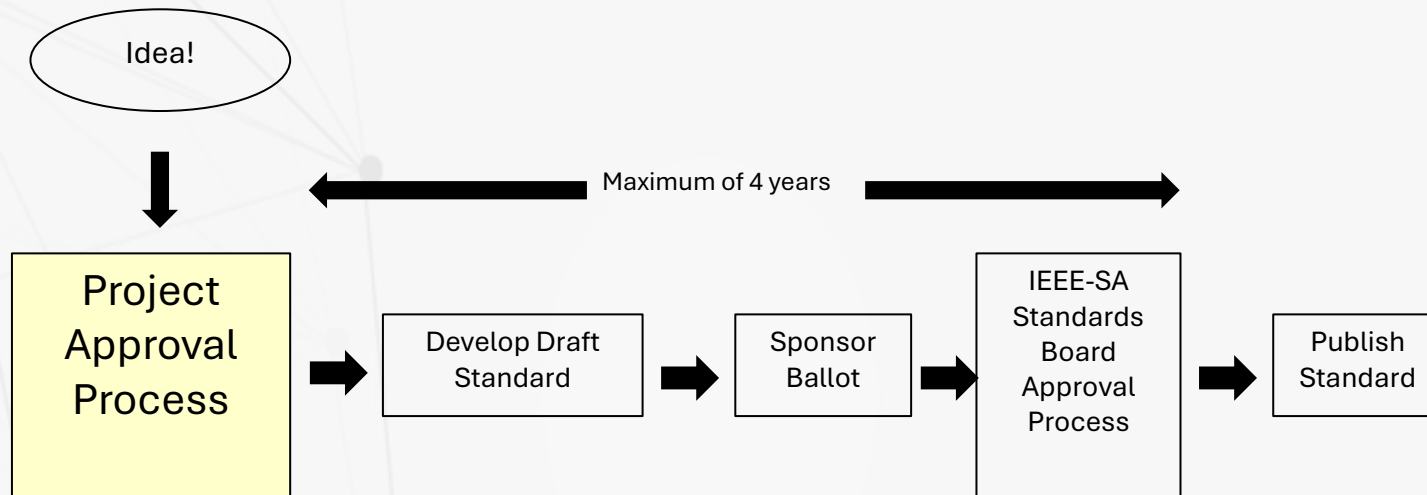
The idea or concept needs to be well defined and accepted for standardization

The Standard is developed under the guidance of a sponsor (e.g., IEEE COM/SDB)



Standardization process

Project authorization



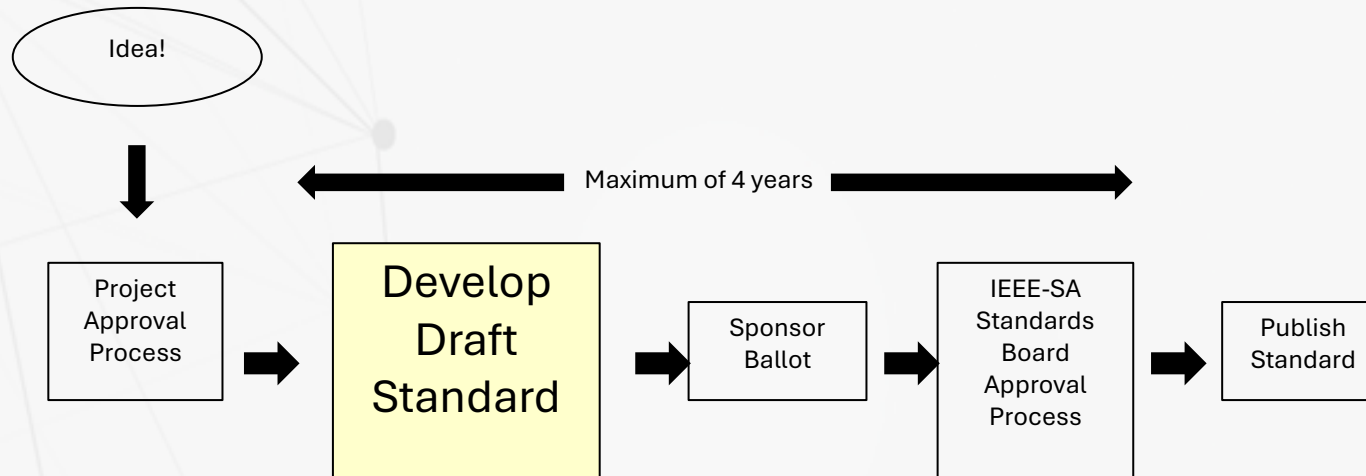
A potential working group or study group gathers to work on the Project Authorization Request (PAR), up to six months before a PAR needs to be submitted.

With the support of the sponsor, submit a PAR to IEEE-SA Standards Board (SASB) for an approval to start the project.

PAR is reviewed by New Standards Committee (NesCom)* and based on its recommendation, IEEE-SA Standards Board (SASB) approves/disapproves the project.

Standardization process

Draft development



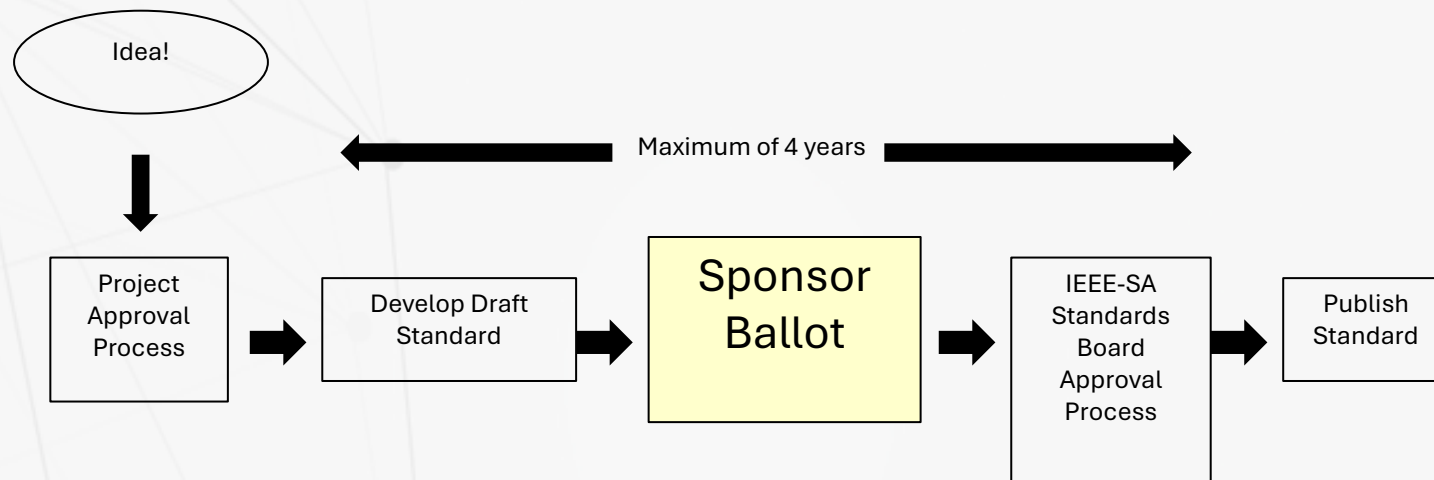
Working group (WG)* is created/maintained under policies and procedures (P&P) of the sponsoring committee (e.g., IEEE P1955 Standard for 6G Empowering Robotics: Use Case Scenarios, Requirements, Architectural Impact, and Technical Assumptions).

WG officers are designated to start the development of the standard. Write drafts of the standard.

Submit finalized draft Mandatory Editorial Coordination (MEC) to ensure conformance with IEEE requirements.

Standardization process

Sponsor balloting



A ballot group is formed using an electronic balloting system called myProject™ . Composition of that balloting group cannot change when the ballot is initiated.

A sponsor ballot is initiated with the draft, to be reviewed, commented, and voted by the ballot group.

- Needs 75% return response rate from the ballot group, and
- Needs 75% affirmative(approved) votes

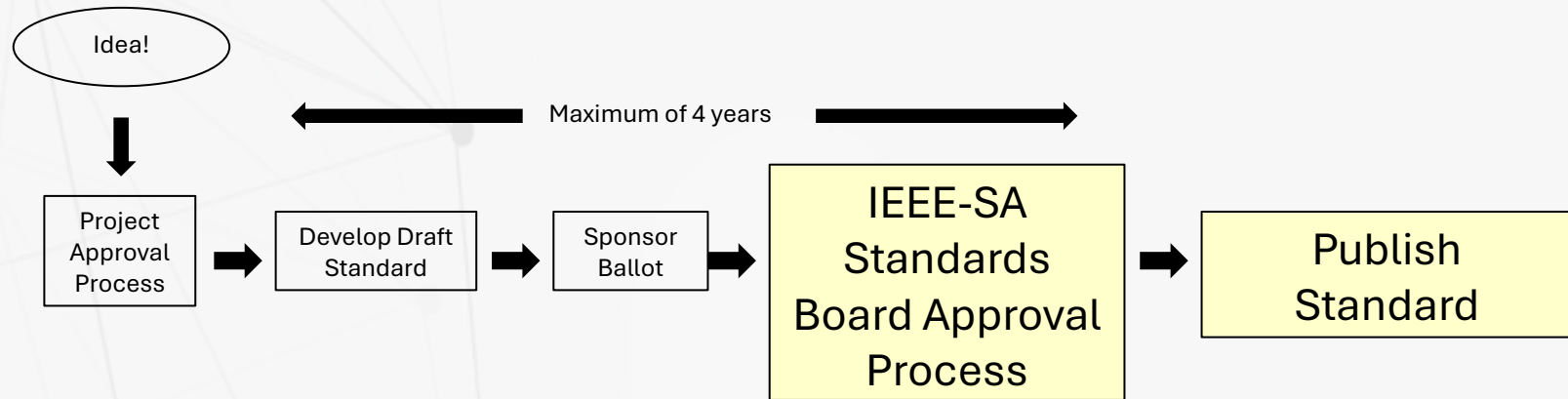
WG reviews all the approved and disapproved votes with comments submitted by the ballot group.

Make a reasonable attempt to resolve all negative votes



Standardization process

Approval process to publication



Submit the final draft standard to Standard Review Committee (RevCom).

RevCom reviews the submitted documents/materials, and makes a recommendation to IEEE-SA Standards Board for an approval of the draft standard.

IEEE-SA Standards Board reviews the recommendation and approve the draft standard.

Publish Standard.

Complimentary copies sent to the WG.

Thank you very much for the careful listening!

Questions?

